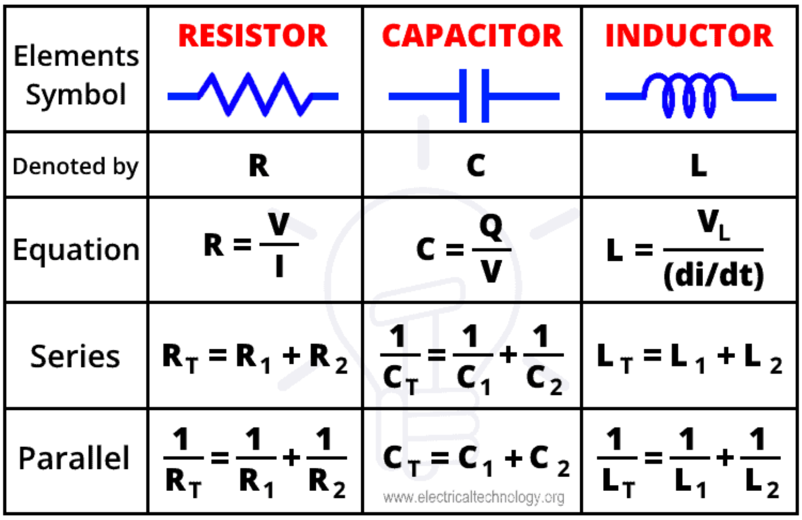
Formula

Basic Electrical Units

Symbol	Quantity	Unit	Symbol
V	Voltage	Volt	**V**
I	Current	Ampere	**A**
R	Resistance	Ohm	**Ω**
P	Power	Watt	**W**
E / W	Energy / Work	Joule	**J**
Q	Electric Charge	Coulomb	**C**
C	Capacitance	Farad	**F**
L	Inductance	Henry	**H**
f	Frequency	Hertz	**Hz**



Elements Symbol	RESISTOR	CAPACITOR	INDUCTOR
Denoted by	R	C	L
Equation	$R = \frac{V}{I}$	$C = \frac{Q}{V}$	$L = \frac{V_L}{(di/dt)}$
Series	$R_T = R_1 + R_2$	$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$	$L_T = L_1 + L_2$
Parallel	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$	$C_T = C_1 + C_2$	$\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2}$

$$R_T = \frac{R_1 R_2}{R_1 + R_1}$$

If
$R = 10\Omega$
$X_L = 10\Omega$
$X_C = 10\Omega$

$Z_R = 10$
$Z_L = j10 \quad / \quad mag\ \angle -90$
$Z_C = -j10 \quad / \quad mag\ \angle 90$

Derived Electrical Units

Symbol	Quantity	Unit	Symbol
G	Conductance	Siemens	**S**
X	Reactance	Ohm	**Ω**
Z	Impedance	Ohm	**Ω**
ϕ	Phase Angle	Degree or Radian	**° / rad**
σ	Conductivity	Siemens per meter	**S/m**

Symbol	Quantity	Unit	Symbol
ρ	Resistivity	Ohm-meter	**Ω·m**

Electromagnetic & Field Units

Symbol	Quantity	Unit	Symbol
B	Magnetic Flux Density	Tesla	**T**
Φ	Magnetic Flux	Weber	**Wb**
H	Magnetic Field Strength	Ampere per meter	**A/m**
μ	Permeability	Henry per meter	**H/m**
ε	Permittivity	Farad per meter	**F/m**
E	Electric Field Strength	Volt per meter	**V/m**

Circuit-Related Units

Symbol	Quantity	Unit	Symbol
t	Time	Second	**s**
T	Period	Second	**s**
λ	Wavelength	Meter	**m**
dB	Decibel (Logarithmic Ratio)	Dimensionless	**dB**

RLC Circuit Formulas

Reactance

$$X_L = \omega L = 2\pi f L$$
$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

Impedance of Series RLC Circuit

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$
$$\theta = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$$

Resonance Frequency

$$f_R = \frac{1}{2\pi\sqrt{LC}}$$

Quality Factor

$$Q = \frac{1}{R} \times \sqrt{\frac{L}{C}}$$
$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 R C}$$

Power in AC Circuits

$$P = VI\cos\theta$$
$$S = VI$$
$$Q = VI\sin\theta$$
$$\cos\theta = \frac{R}{Z}$$

Resonance

In a **Series RLC circuit** (which contains a **Resistor (R)**, **Inductor (L)**, and **Capacitor (C)** in series), **resonance** occurs when the **inductive reactance** and **capacitive reactance** are equal in magnitude but opposite in sign:

$$X_L = X_C \quad \text{where} \quad X_L = 2\pi f L \quad \text{and} \quad X_C = \frac{1}{2\pi f C}$$

At this point:

- The total **reactance** $X = X_L - X_C = 0$
- The **impedance** of the circuit is purely **resistive**: $Z = R$
- The **current** is at its **maximum** for a given voltage
- The **voltage** across L and C can be large, but they cancel each other out in terms of phase

The **resonant frequency** f_0 is given by:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

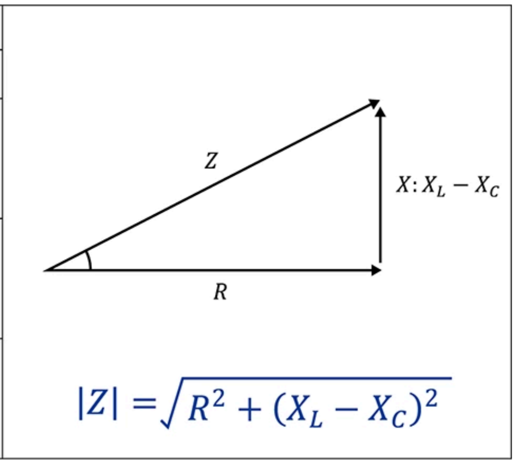
So, if the circuit is "**in resonance**," it simply means it's operating at this special frequency f_0, where the energy oscillates efficiently between the inductor and capacitor.

Impedance

Impedance, represented by the symbol Z, is **a measure of the opposition to electrical flow**. It is measured in ohms (Ω).

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Circuit element	Impedance	
	Symbol	Value
Resistor	R	R
Capacitor	X_C	$\frac{1}{j\omega C}$
Inductor	X_L	$j\omega L$



DC-DC Converter Formulas

Buck Converter (Step-Down)

$$V_{out} = DV_{in}$$
$$I_L = \frac{V_{in} - V_{out}}{L} \cdot DT$$
$$I_{out} = I_{in} \cdot D$$

Boost Converter (Step-Up)

$$V_{out} = \frac{V_{in}}{1 - D}$$
$$I_{out} = I_{in} \cdot (1 - D)$$

Buck-Boost Converter

$$V_{out} = \frac{D}{1 - D} V_{in}$$

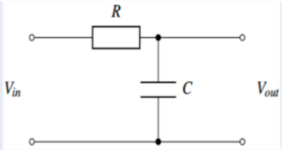
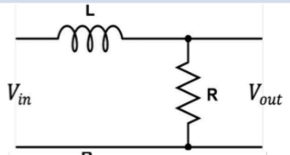
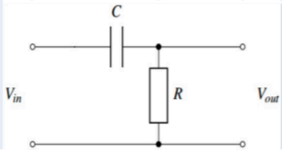
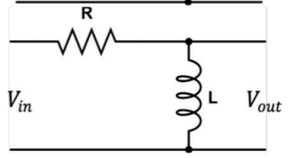
Inductor Current Ripple

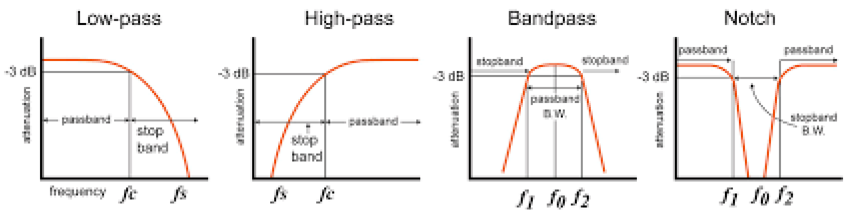
$$\Delta I_L = \frac{V_{in} D}{L f_s}$$

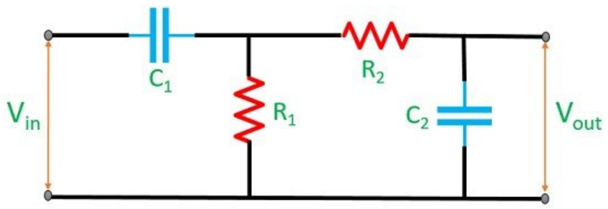
Capacitor Voltage Ripple

$$\Delta V_C = \frac{I_{out} D}{C f_s}$$

Filter

	RC			RL			Property
	G	f_{cutoff}	circuit	G	f_{cutoff}	circuit	
Low pass	$\frac{1}{1 + j\omega RC}$	$\frac{1}{2\pi RC}$		$\frac{1}{1 + j\omega \frac{L}{R}}$	$\frac{R}{2\pi L}$		Attenuate high frequency signals
High pass	$\frac{\omega RC}{\sqrt{1 + (\omega RC)^2}}$			$\frac{j\omega \frac{L}{R}}{1 + j\omega \frac{L}{R}}$			Attenuate low frequency signals





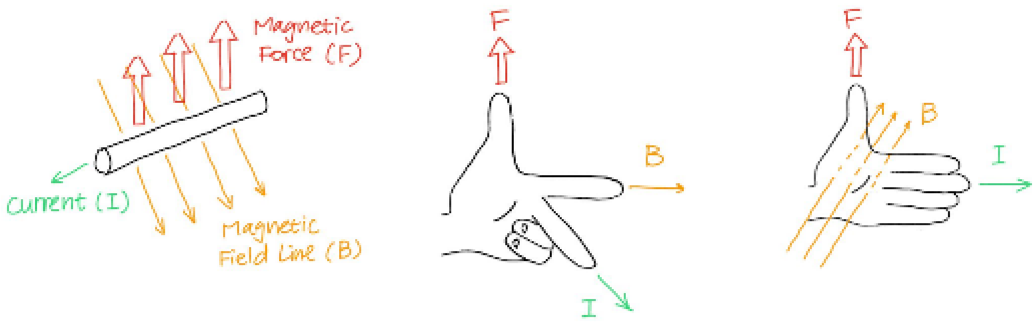
Circuit diagram of Bandpass filter

Electronics Coach

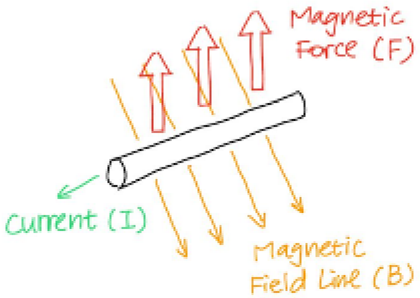
Motor

Force acting on Current-carrying Wire in a Magnetic Field

When current (I) flows through a wire in a magnetic field (B), the charges experience the magnetic force (F), and the entire wire is pushed in a direction perpendicular to the plane containing the B vector.



We use left-hand rules to remember the direction of the force. As illustrated, they both work, use the one familiar to you.



The magnitude of the force F acting on the wire is:

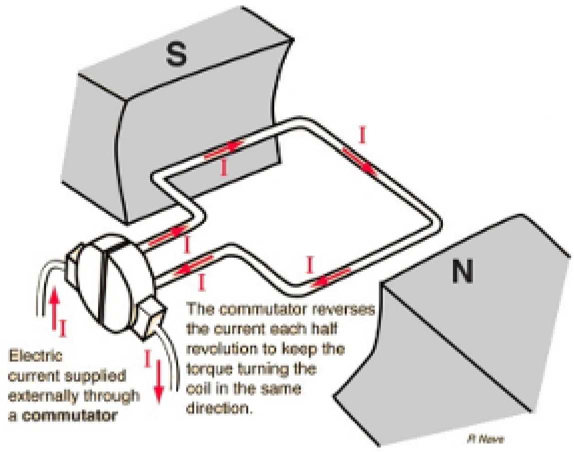
$$F = I\ell B \sin\theta$$

where θ is the angle between the B vector and the direction of current I, and ℓ is the length of the wire.

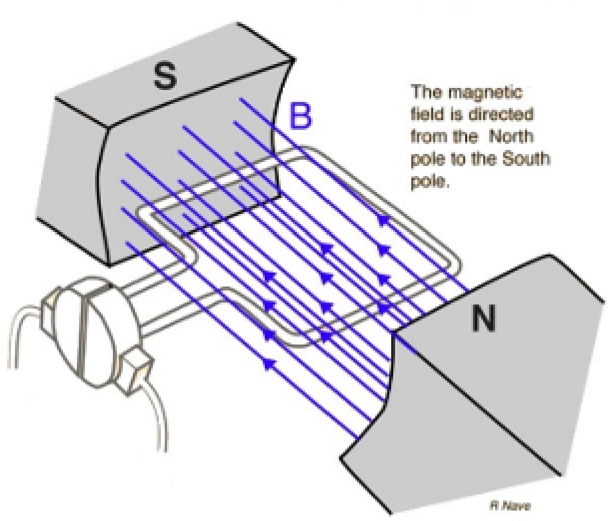
How is Torque produced in a DC Motor?

In a DC motor, wires are bent to form rectangular loops. The image below shows one such loop.

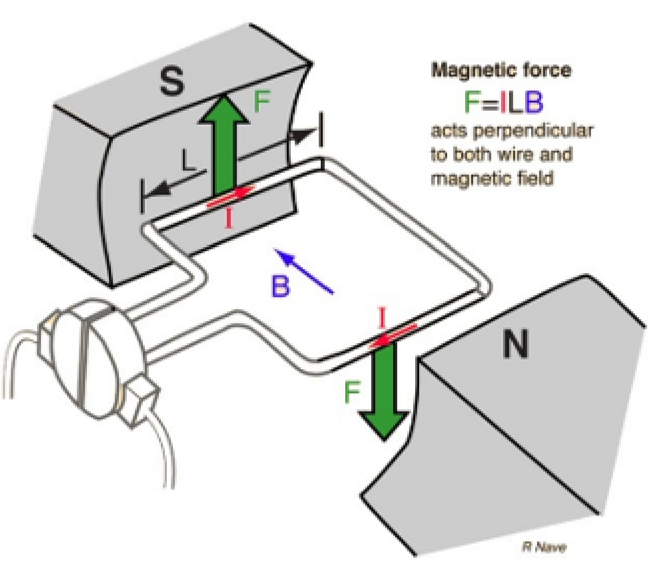
Current (I) flow in coil

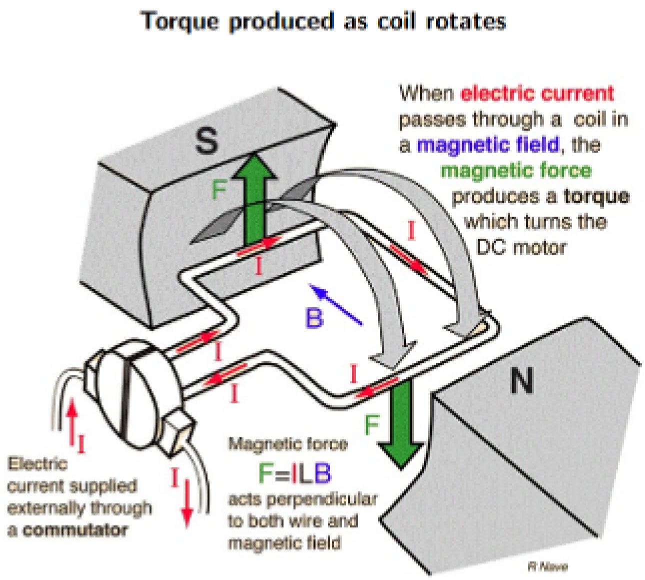


Magnetic field (B) between poles

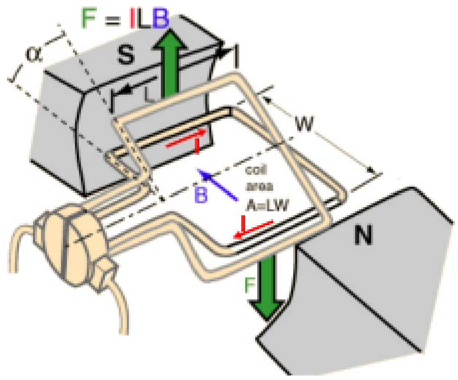


Magnetic force (F) acting on coil



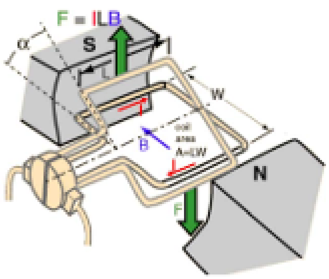


Two sides of the coil experience force in opposite directions creating a couple.



The **moment** of the couple is FW where the **moment arm** W is the distance between the two conductors experiencing the magnetic force.

We consider the case when the segments of the loop experiencing the magnetic force are **perpendicular** to the B field, as shown:



The moment is:

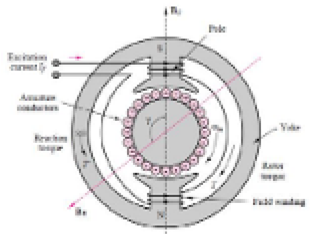
$$FW = I\ell BW$$

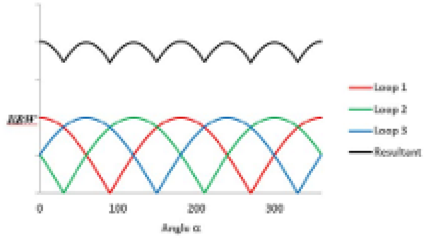
The moment arm, W, changes when the loop rotates. It is maximum when $\alpha = 0^\circ$ and becomes zero when $\alpha = 90^\circ$.

So, the moment ($I\ell BW$) produced by a single loop **varies as it rotates**.

Moment of Multiple Coiled Coils

So far, we looked at a single coil. In an actual DC motor, multiple loops are wound around a magnetic core.





Though the moment of one loop varies, the cumulative effect of the moments from all loops remains nearly constant.

Figure on the right shows the variation of the moments of three different loops which are placed at 120° apart from one another. The sum of the three moments is plotted in black.

Imagine if there exist n pairs of conductors, the resultant moment will be smoothed out.

Electromagnetic Torque

The sum of moments of all loops is the **electromagnetic torque** produced by a motor. Which has a nearly constant value with negligibly small ripples.

If there are n pairs of the conductors, then the electromagnetic torque is:

$$T_{em} = nI\ell BW$$

Where:

- n: No. of loops/ conductor pairs.
- I: current supplied.
- ℓ: length of each conductor.
- W: (maximum) distance between the two conductors.
- B: magnetic flux density.

For a PMDC motor that is already assembled, the parameters $n, I, B, andW$ are fixed quantities.

So, in the equation:

$$T_{em} = nI\ell BW$$

the product $n\ell BW$ is a constant, it is called the **torque constant**, K_t, of the motor.

Hence the electromagnetic torque can be represented as:

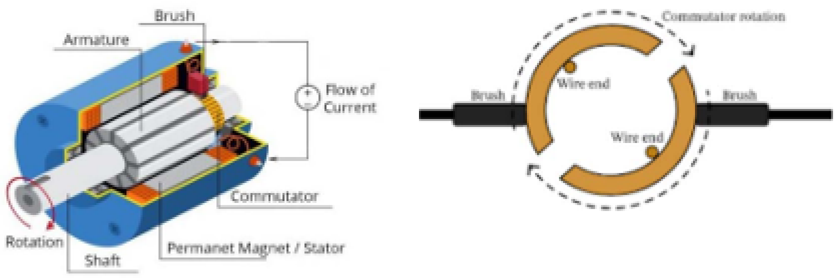
$$T_{em} = K_t I$$

The unit of K_t is newton-meter per ampere ($\frac{Nm}{A}$)

We refer to the current, I, as the **rotor current** or **armature current**, denoted as I_a.

DC Motor Internal Construction

Let's look at a more realistic illustration of a DC motor:



- **Sactor**: the permanent magnets are fixed on the stator. The magnetic field produced by the stator is also known as the **stator field**
- **Armature/ Rotor**: The center core rotates under the influence of the torque, This rotating part of is called the **rotor** or **armature**.
- **Commutator**: The external source supplies current to the armature through the brush, where the conductive commutator is fixed with the coils.

Exciting the DC Motor

When you connect a DC motor to the power supply, it **does not spin with a steady speed instantaneously**, rather it takes some time.

The armature coil is both **resistive** and **inductive**. (Why?)

Inductance does not allow instantaneous change of current.

So, when a DC voltage is applied to the armature coil, the armature current grows gradually to the steady state value with the **electrical time constant**:

$$\tau_e = \frac{L_a}{R_a}$$

Where L_a and R_a are the inductance and resistance, respectively, of the armature coil.

Exciting the DC Motor for sufficient time

During the transient state, the motor speed keeps increasing. If we wait for long enough, what will happen?

Newton's Second Law states:

The change of motion of an object is proportional to the force impressed.

$$F = ma$$

Known from the equation:

$$F = I\ell B$$

the magnetic force is always acting on the armature, as long as there is current flowing in the coils.
As the mass of the armature is fixed, and $F \neq 0$, acceleration $a \neq 0$ either.

That is to say, if we supply a DC voltage to a motor, and let it run long enough, the motor speed increases indefinitely!

Limiting Factor of the Motor Speed

In reality, we do not see a motor speed increase indefinitely, at a certain point, the speed reaches a steady state.
Two factors constrain the speed:

- Friction (brush, bearing, air, etc.)
- Back EMF

Motor Speed

The unit of velocity we used in other cases is usually $\frac{rad}{s}$.

A more commonly used unit of motor speed is **revolution per minute** or **RPM** which is related to ω by the formula:

$$\omega = 2\pi \frac{RPM}{60}$$

Friction

As friction exists, it will oppose the motion of the armature.
That causes the net torque acting on the rotor to be:

$$T_{net} = T_{em} - T_f$$

Where:

- T_{em} is the electromagnet electromagnetic torque
- Tf is the **friction torque**

The friction torque, T_f is proportional to the motor velocity:

$$T_f = K_f \cdot \omega$$

Approaching a steady state, when the motor is not loaded, an equilibrium between T_{em} is T_f is reached:

$$T_{em} = T_f$$
$$T_{net} = T_{em} - T_f = 0$$
$$\omega_{ss} = \frac{T_f}{K_f} = \frac{T_{em}}{K_f}$$

Fraction limits the motor speed **upper limit** by bounding the net torque.

Back EMF

The friction does not let the speed go unbounded. There is one more phenomenon that gives the DC motor a self-regulating property.

Earlier we considered the charges moving **in the conductor** under the influence of an external source.

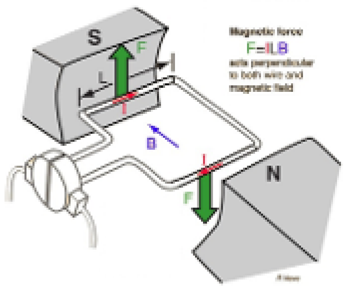
Lorentz force Law states:

A charge moving in a magnetic field experiences a force perpendicular to the plane containing the direction of the magnetic field and the velocity of the moving charge.

This is why the coils rotate and produce EM torque.

Meanwhile, as the coil moves, it experiences a **change** in the magnetic field, which causes an **EMF to be induced** in the coil.

The induced voltage **opposes** the applied voltage that sets the rotor current to produce the torque.



This induced voltage is called the **back electromotive force**, in short, **back EMF**.

Laws of electromagnetism for your reference:

Faraday's law states:

In a closed coil, a change in the magnetic environment of the coil causes a **voltage**, *or* **EMF** *(electromotive force), to be induced in the coil.*

Lenz's law states:

When EMF is induced by a change in magnetic flux, the polarity of the induced EMF is such that it generates a current whose magnetic field is in a direction that **opposes the change** *that produced it (the original magnetic field).*

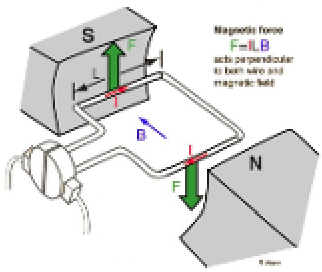
Likewise, the faster the rotor rotates, the higher the induced voltage. Back EMF is proportional to the angular velocity:

$$V_e = K_e \cdot \omega$$

Where:

- K_e is the **back EMF constant**, unit: $\frac{V}{rad/s}$ or $\frac{Vs}{rad}$
- For a PMDC motor, the numerical values of **torque constant**, K_t, and **back EMF constant**, K_e are equal.

The induced back EMF also causes a "back current", that reduces the overall current flows in the coil.
(There isn't an actual current that is flowing against the supplied current!)



The magnetic force acting on the coil is given by:

$$F = I\ell B$$

When I is reduced, F is also reduced, hence the acceleration of armature is reduced.

Back EMF limits the motor speed at any time by offsetting the supplied voltage partially, hence reducing the current flowing in the coil.

Summary

- When a current flows in the coil, due to the presence of a magnetic field, a magnetic force is acting on the coil and drives the armature to rotate.
- During transient: the motor speed gradually builds up, as the inductance of the coil does not allow instantaneous change of current.
- Friction torque T_f and back EMF V_e opposes the armature rotation, causing the speed of the motor to be regulated.

i. Speed is always proportional to the back emf ($\omega \propto Eb$)
ii. Electromagnetic torque developed is always proportional to armature current ($Tem \propto Iarmature$)
iii. Whenever load torque is increased, speed is decreased.

Characteristic Equation of PMDC Motor

We will now move on to characterize the motor Substituting $I_a = \frac{T_{em}}{K_t}$ and $V_e = K_e\omega$ in the following equation:

$$V_e = V_s - I_a R_a$$

obtains:

$$\omega = \frac{V_s}{K_e} - \frac{R_a}{K_t K_e} T_{em}$$

This relation $\omega = f(T_{em})$ is the equation of a straight line with a negative slope of $-\frac{R}{K_t K_e}$.

This is one of the most important equations for this topic.

Question 2

Torque constant and armature resistance of a PMDC motor are 50 mNm/A and 10 Ω, respectively. The motor is driven by a 10V DC source.

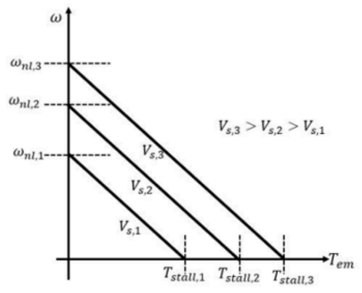
1. What is the maximum mechanical power this motor can produce?
2. What are the torque and the speed at the maximum power?

The maximum power developed by the motor is at half of stall torque and half of no-load speed.

$$\begin{aligned}
\omega_{nl} &= \frac{V_s}{K_e} = \frac{V_s}{K_t} \\
&= \frac{10V}{50mNm/A} = \frac{10V}{50mV/rad/s} = 200rad/s \\
T_{stall} &= K_t \frac{V_s}{R_a} \\
&= 50\frac{mNm}{A} \times \frac{10V}{50\Omega} = 50mNm \\
P_{max} &= \frac{T_{stall}}{2} \times \frac{\omega_{nl}}{2} = 2.5W
\end{aligned}$$

Speed versus Torque Graph

The speed versus torque graph of a PMDC motor differs with different applied voltages, V_S.



If voltage is changed, no-load speed and stalled torque will also change accordingly.

- Suppose it is required to maintain the speed while changing the required torque, we need to changing the voltage accordingly.

With different values of V_s, the graph will be shifted up or down keeping the slope unchanged.

Operating Point

The equation below gives all possible combinations of speed and torque produced for a given V_s:

$$\omega = \frac{V_s}{K_e} - \frac{R_a}{K_t K_e} T_{em}$$

To determine the exact motor speed and torque produced, we need to analyze the equilibrium of torques acting on the system.

At steady state, net torque must be zero.

No Load attached to the Motor Shaft

Earlier we discussed the theoretical steady-state speed when the motor produces no torque, and we know that this will never happen in real life as friction exists.

A small EM torque (T_{em}) is produced to overcome the friction torque (T_f) so that the **net torque is zero**.

$$T_{em} = T_f$$

Friction torque T_f is propotional to speed:

$$T_f = K_f \cdot \omega$$

From the above two equations, we derive:

$$\omega = \frac{T_{em}}{K_f}$$

Mechanical load attached to the Motor Shaft

In this condition, the motor must produce enough torque to overcome the load torque (T_L) as well as the friction torque (T_f) to accomplish the intended task.
Notice that the faction here not only coming from the bearing but also from the brush, air, etc.

To achieve equilibrium:

$$T_{net} = T_{em} - T_f - T_L = 0$$

implies that:

$$T_{em} = T_f + T_L$$

We have seen that T_f varies with speed. Depending on the type of task, the load torque (T_L) may or may not vary with speed. Hence, it would be challenging to describe the relationship between operating speed and the torques with a straightforward equation.

If we plot the ω versus $T_f + T_L$ graph on the same plot showing ω versus T_{em}, then the point of intersection between the two graphs gives the operating point.

Question 3

Suppose, while the object is being lifted, another object falls on the box. The pulley system continues to pull the box and the new object upward.

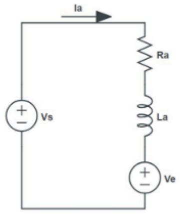
1. If the voltage applied to the motor is not changed, how will the lifting speed be affected?
2. What should be done to keep the speed unchanged?
3. If the string between the pulley and the object is cut, the object will drop to the ground. What will happen to the motor?

1. The required torque is increased the motor will produce that at the expense of the speed.
2. If it is required to maintain the speed when the required torque is changed, the applied voltage must be changed so that the speed-torque graph is shifted upward/ downward.
3. The required torque is decreased, and the speed will go up.

Power and Efficiency

A motor converts electrical energy to mechanical energy. Let's track from the power supplied to the motor to the power output by the motor:

- **Input electrical power**:
$$P_{in} = V_S I_a$$
- Power loss in the rotor resistance, **copper loss**
$$P_{cu\_loss} = I_a^2 R_a$$
- Input power minus copper loss is the **power developed** by the motor, P_{dev}



$$P_{dev} = P_{in} - P_{cu\_loss}$$
$$= V_S I_a - I_a^2 R_a = (V_S - I_a R_a) I_a = V_e I_a$$

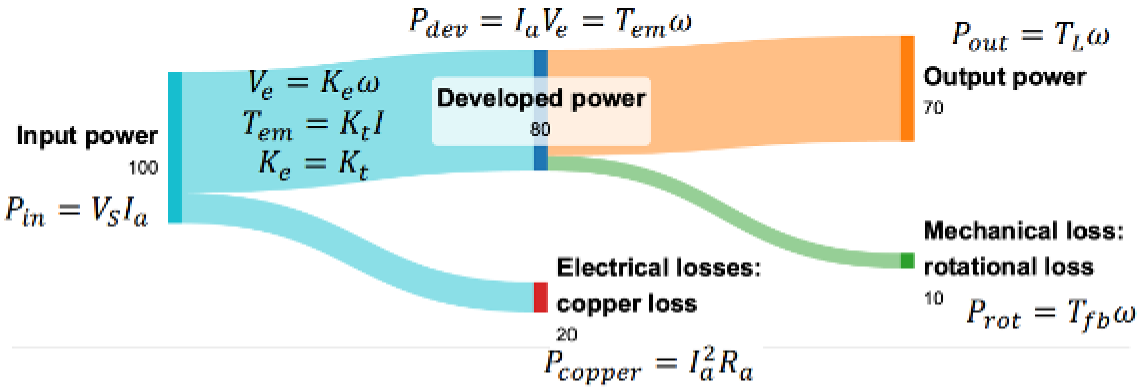
- **Output power** P_{out} is the product of output torque T_L, applied to the load and the angular speed ω
$$P_{out} = \omega T_L$$
- The motor **efficiency**: $\eta = \frac{P_{out}}{P_{in}}$

- Developed power: power available for doing mechanical work

$$P_{dev} = P_{in} - P_{copper}$$
$$P_{dev} = V_S I_a - I_a^2 R_a$$
$$P_{dev} = I_a V_e = T_{em}\omega$$

Power of a torque:
$P = T\omega$

- In practice, we can use Sankey diagram to denote the power flow of a system. For a PMDC motor:



- From the Sankey diagram, the power flow of our simplified PMDC motor is:

$$P_{out} = P_{in} - P_{copper} - P_{rot}$$

- Overall, the efficiency of a DC motor is

$$\eta = \frac{P_{out}}{P_{in}} = \frac{T_L \omega}{V_S I}$$

Question 4

PMDC motor draws 0.8A current when connected to a 12V DC source and spins at 3600 RPM. If the efficiency of the motor is 80%, how much torque is applied at the output shaft?

The input power:

$$P_{in} = V_S I_a = 12V \times 0.8A = 9.6W$$

The output power:

$$P_{out} = \eta \times P_{in} = 0.8 \times 9.6W = 7.68W$$

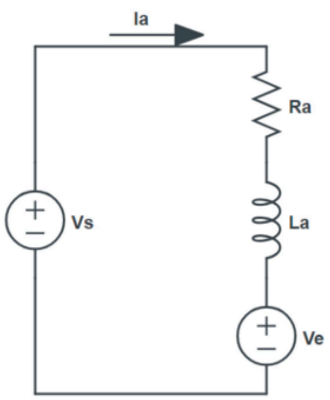
The speed:

$$\omega = 2\pi \frac{RPM}{60} = 2\pi \frac{3600}{60} = 377 rad/s$$

Torque at the output shaft:

$$T_L = \frac{P_{out}}{\omega} = \frac{7.68W}{377 rad/s} = 20mNm$$

Shaft Torque



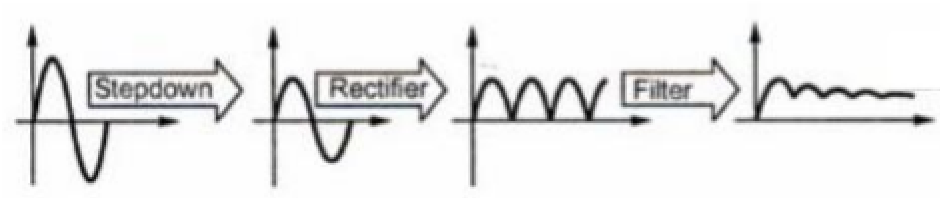
- Rotational loss (P_{rot}): the power loss to overcome friction on the shaft.
- Load torque (T_L): shaft output torque applied to load

$$P_{out} = P_{dev} - P_{rot}$$
$$T_L \omega = T_{em} \omega - T_{fb} \omega$$
$$T_L = T_{em} - T_{fb}$$

Power of a torque:
$P = T\omega$

DC Supply

- **Change in amplitude (AC $\rightarrow$ AC)**
  - A *transformer* steps down utility grid 230V AC to lower voltage AC.
- **Controlling polarity (AC $\rightarrow$ DC)**
  - A *rectifier* fixes polarity of AC voltage.
- **Making the DC voltage smoother (DC $\rightarrow$ DC)**
  - A capacitor used as a *filter* helps produce a nearly constant DC voltage.



Electricity Transmission at High Voltage

Resistance of the transmission cables will have power losses equal to

$$I^2 R_{line}$$

To reduce power loss:

- Decrease the resistance (costly and impractical)
- Decrease magnitude of the current (How?)

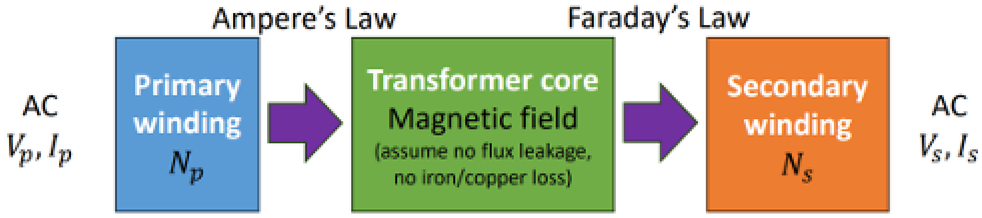
The required power to be transmitted is:

$$P = V \times I$$

To keep P constant, we choose to increase V hence I will be reduced. Thus, $I^2 R_{line}$ is reduced.

At the generator, a **step-up transformer** increases the voltage level. At the consumer end, the voltage is reduced to a safer level with **step-down transformers**. In practice, this is usually done in several stages.

Ideal Transformer



- N_p turns of primary winding, N_s turns of secondary windings

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Turns-ratio

- Since ideal transformers are lossless, $V_p I_p = V_s I_s$

$$\frac{I_s}{I_p} = \frac{N_p}{N_s}$$

The current will flow if a load is connected to the secondary side winding.

Consider an ideal case with no power loss in the winding or the core, the power will be fully transferred from the primary side to the secondary side:

$$V_p \times I_p = V_s \times I_s$$

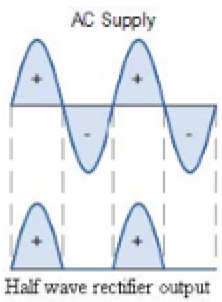
$$\frac{V_s}{V_p} = \frac{I_p}{I_s}$$

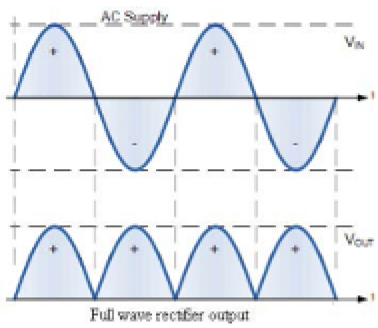
$$\frac{I_s}{I_p} = \frac{N_p}{N_s}$$

AC-DC Rectifier

A rectifier converts the AC voltage to the DC voltage. The rectifier keeps the polarity on the DC side fixed by doing one of the following:

- Allow power to flow to the load when AC voltage has **one** polarity, and block power flow when the polarity is opposite. (Half-wave rectifier)
- Allow power to flow to the load during **both** the positive and negative half of the AC voltage but channel the current in the same direction through the load. (Full wave rectifier)



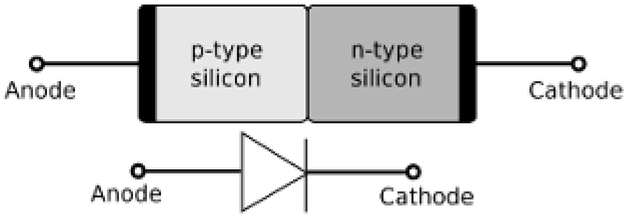


Diode

The diode is an essential component in rectifiers.

Including LED, all diodes are semiconductor devices with a PN junction.

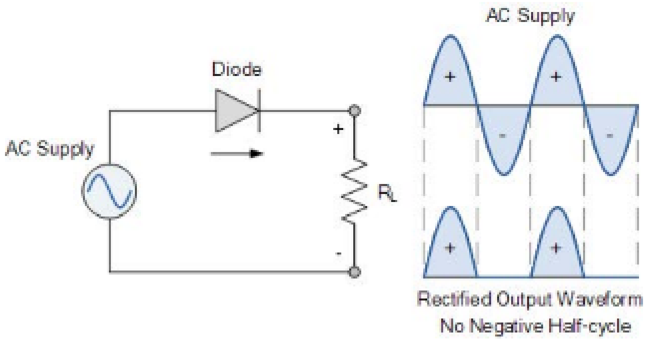
When the diode is **forward biased**, the voltage applied at the P-side is higher, and current flows from P to N.
When the diode is **reverse biased**, the voltage applied at the P-side is lower, and negligible current flows from N to P, we treat it as an open circuit.



In the diode symbol, you can treat the triangle as an indicator of the direction of current flow.

Half Wave Rectifier (HWR)

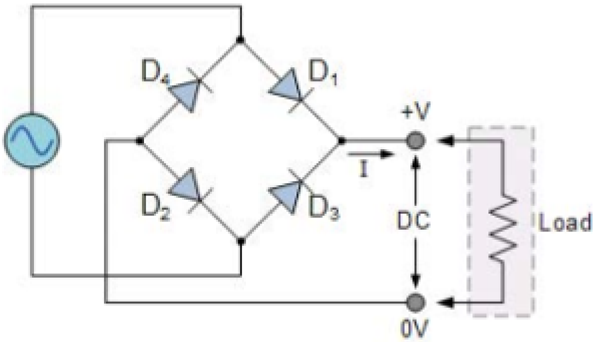
A single diode serves as a half-wave rectifier:



The load does not receive power when the diode is reverse biased. The rectification efficiency is low as power is delivered to the load during half-cycle only.

Full Wave Rectifier (FWR)

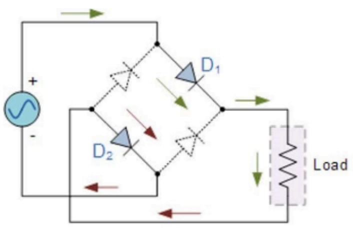
A full wave rectifier significantly increases the rectification efficiency. In this course, we will focus on only **diode bridge rectifier**.



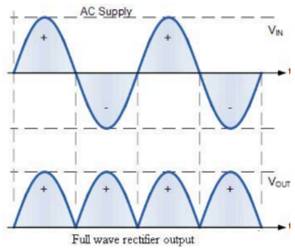
When the load is connected, current will flow in both cycles of the AC voltage. However, in both cycles, the flow of current is in the same direction as shown by the arrow.

During the **positive** half-cycle:

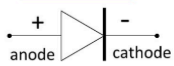
- Diodes D1 and D2 conduct.
- AC supply is directly connected to the load. (+ to +, - to -)



D1 and D2 are forward biased
D3 and D4 are reverse biased

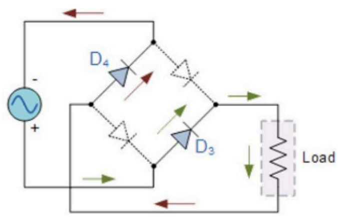


Direction of Current Conduction

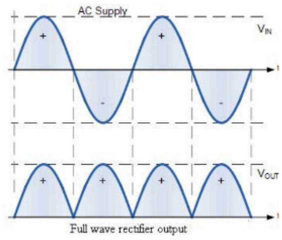


During the **negative** half-cycle:

- Diodes D3 and D4 conduct.
- effectively swapping the terminals of the AC supply and hence the load sees a positive voltage. (+ to -, - to +)



D1 and D2 are reverse biased
D3 and D4 are forward biased



Let the AC voltage be

$$v_{in}(t) = V_m \sin \omega t$$

The voltage seen at the load is DC with a ripple of amplitude equal to V_m, while its average voltage is:

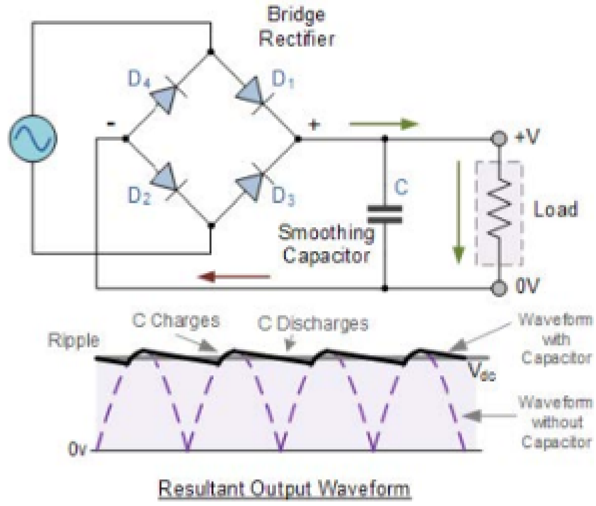
$$V_{o,avg} = \frac{2V_m}{\pi} \approx 0.63 V_m$$

Average voltage:

- $v_{o,avg} = \frac{1}{T/2}\int_0^{T/2} V_m \sin \omega t \, dt$
- $v_{o,avg} = \frac{1}{T/2}\frac{1}{\omega}\left[-V_m \cos \omega t\right]_0^{T/2}$
- $v_{o,avg} = \frac{2V_m}{\pi} \approx 0.637 V_m$

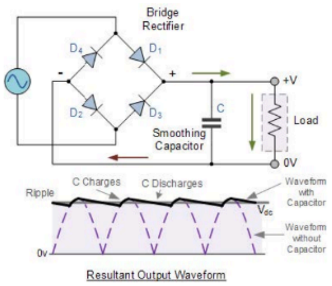
Diode Bridge Rectifier with Capacitor Filter

The large ripple at the rectifier output can be minimized by adding a capacitor as a filter.



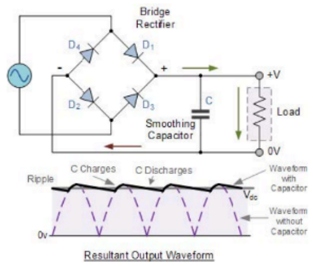
When the input AC voltage is higher than the capacitor voltage:

- one pair of diodes conduct
- load receives power directly from the AC supply
- the capacitor gets charged simultaneously

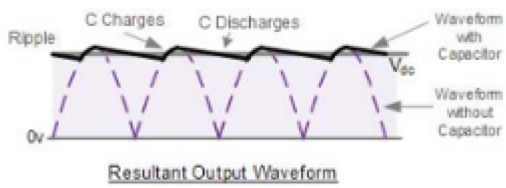


When the input AC voltage is lower than the capacitor voltage:

- the diodes stop conducting
- the load resistor gets power from the energy stored in the capacitor
- capacitor discharges and its voltage decreases



Fluctuation is still present in the output voltage, but the ripple is much smaller than that seen without the capacitor.



The magnitude of the *peak-t-peak ripple voltage* (ΔV) depends on the values of the capacitance (C), the load resistance (R), the output voltage level (V), and the frequency of the AC supply (f):

$$\Delta V = \frac{V}{2fRC}$$

We usually choose a large enough C so that the capacitor discharges slowly, and it takes approximately $\frac{T}{2}$ for V_C to fall below the AC voltage.